

Written Rheumatology

1. Rheumatoid Arthritis (RA) :-

1. ACR/EULAR Criteria for the diagnosis of Rheumatoid Arthritis.

(RA أعلى من أو يساوي 6 يبقى كدة score ، اللي هو لو الـ scoring ، و متتساش نكتب الـ domains بنكتب الـ)

Domains		Score
A Joint involvement	1 large joint	0
	2-10 large joints	1
	1-3 small joints	2
	4-10 small joints	3
	>10 joints	5
B Serology	Negative RF and negative ACPA	0
	Low-positive RF or low-positive ACPA	2
	High-positive RF or high-positive ACPA	3
C Acute-phase reactants	Normal CRP and normal ESR	0
	Abnormal CRP or Abnormal ESR	1
D Duration of symptoms	<6 weeks	0
	≥6 weeks	1

Patients with a score ≥6 are classified definite RA.

2. Management of Rheumatoid Arthritis (investigation, assessment & treatment).

(pharmacological بس و بالاختصاص الـ +++ او ممكن تطلب الـ)

Laboratory tests

1) Rheumatoid Factor (RF):

- present in 70 - 80% of RA patients.
- Not specific to RA due to Found in other conditions like: old age - HCV - cryoglobulinemia - Graves' disease - Sjögren's syndrome.

2) Anti-CCP Antibodies (ACPA):

- Found in 60-70% of RA patients.
- Highly specific for RA.
- Both RF and ACPA presence associated with more severe disease course and important for diagnosis and for staging the disease.

Inflammatory markers:

- 1- elevated ESR and CRP
- 2- Thrombocytosis and leukocytosis in active disease.
- 3- reduced hemoglobin is also common
- 4- serum iron may be lowered whereas ferritin concentration may be elevated as a reflection of acute phase reaction
- 5- liver function test and creatinine as well as urine analysis to assess organ dysfunction



Imaging Procedures

- Early detection of joint damage is essential for timely intervention & better outcomes.

1) Role of Ultrasonography:

- Detection of synovitis in involved joints.
- Identification of subclinical synovitis.
- Monitoring treatment response.
- Guidance for joint aspirations and injections.

2) Role of Doppler Ultrasonography:

- Detects increased vascularity in inflamed synovium.
- Helps grade disease activity (Grades 0-3).
- Predicts erosive disease and monitors treatment response. → used in Follow up.

Conventional Radiography (X-rays) :

- gold standard for diagnosing joint damage.
- Limitations → Early inflammatory changes, soft tissue involvement may not be visible.

Treatment:

1. General Principles

- **Early Intervention:** Start Disease-Modifying Antirheumatic Drugs (DMARDs) as soon as the diagnosis is made (the "window of opportunity").
- **Frequent Monitoring:** Assess disease activity (using scores like DAS28, SDAI, or CDAI) every 1-3 months and adjust therapy if the target isn't met.

2. Pharmacological Therapy

A. Conventional Synthetic DMARDs

These are the first line of defense.

- **Methotrexate (MTX):** The "Gold Standard" and anchor drug. Usually started at 7.5-15 mg/ week with folic acid supplementation.
- **Leflunomide:** An alternative if MTX is contraindicated or not tolerated.
- **Sulfasalazine & Hydroxychloroquine:**

B. Glucocorticoids (Bridging Therapy)

- Used at low doses for the shortest time possible, in early disease or during flare up
- Purpose: To control inflammation (bridging therapy) quickly while waiting for csDMARDs (which take 6-12 weeks) to kick in. (intra-articular)

C. Biological DMARDs

If csDMARDs fail after 3-6 months, if there are contraindications of MTX poor prognostic factors like high RF/ACPA, early erosion, high activity.

- **TNF Inhibitors:** (e.g., Adalimumab, Etanercept, Infliximab).
- **IL-6 Inhibitors:** (e.g., Tocilizumab).
- **B-cell Depletion:** (e.g., Rituximab)
- **T-cell Co-stimulation Blockers:** (e.g., Abatacept).

D. Targeted Synthetic DMARDs :

- **JAK Inhibitors:** (e.g., Tofacitinib, Baricitinib).



3. Non-Pharmacological treatment

- **Patient Education:** Understanding the chronic nature of the disease and drug side effects.
- **Physical Therapy (PT):** To maintain joint range of motion and muscle strength (improve).
- **Occupational Therapy (OT):** For joint protection techniques and splints.
- **Cardiovascular Risk Management:** RA patients have a higher risk of IHD; managing BP and lipids is crucial.
- **Osteoporosis Prevention:** Especially if the patient is on long-term steroids.

4. Surgical Intervention

Reserved for end-stage joint destruction or complications:

- **Synovectomy:** If medical management fails to control localized synovitis.
- **Arthroplasty (Joint Replacement):** Most commonly Hip or Knee, to restore function and relieve pain.
- **Tendon Repair:** For ruptured tendons (common in the RA hand)

Disease Assessment and Follow-up

Disease activity assessment is crucial for treatment modification.

1. **Disease Activity Score (DAS 28)** Used to assess disease activity in rheumatoid arthritis depending on tender and swollen joint count, inflammatory markers (CRP or ESR), and patient global assessment.
2. **Health Assessment Questionnaire (HAQ score)** Used to evaluate functional disability in rheumatoid arthritis by assessing the patient's ability to perform daily activities

2. Spondarthropathy (SPA) :-

3. SPA subgroups.

1- Axial Spondarthropathy (axSpA):

- Radiographic axSpA: Also known as classic Ankylosing Spondylitis (AS) .
- Non-radiographic axSpA (nr-axSpA)

2- Peripheral Spondarthropathy pSpA) :

- ankylosing spondylitis
- acute anterior uveitis
- Reactive arthritis
- Psoriatic arthritis
- juvenile SPA
- undifferentiated SPA
- enteropathic SPA



4. Rheumatological (articular) manifestations.

1) Inflammatory Back Pain (IBP)

- IBP is the first manifestation in 75% of patients.
- Classically, pain starts in lumbar region or at lumbosacral junction, Sacroiliac joint.
- Typically a dull pain of insidious onset, becoming persistent after few months.
- It is inflammatory: Pain worsens with inactivity and improves with exercise & NSAIDs.
- Morning stiffness is often prolonged (> 30 minutes) and nocturnal pain may awaken patients from sleep.
- Radiographic Changes: Syndesmophytes in lumbar spine

2) Sacroiliitis: Must do MRI to diagnosis: shows Bone marrow edema

3) Fracture

- Spinal osteoporosis is commonly seen in longstanding SpA resulting in increased fracture risk.
- This contributes to the high prevalence of fractures which often occurs after very minimal trauma to the rigid, ankylosed spine (Fusion & syndesmophyte).
- Spinal osteoporosis is partly due to the lack of mobility as a consequence of the disease, as well as a result of pro-inflammatory cytokines (TNF- α).

4) Peripheral Arthritis

- Typically asymmetric, oligoarticular and involves the lower limbs (e.g., Ankle, Knee).
- Upper limb involvement is typically associated with PsA (Psoriatic arthritis).
- A bilateral symmetrical polyarticular presentation is possible, which differs from rheumatoid arthritis in that the distal interphalangeal joints (DIP) are often affected. (Destruction + new bone formation)
- Arthritis of hips & shoulders often occurs early in first 10 years of disease.
- Hip involvement is commonly bilateral, except enteropathic.

5) Dactylitis (Sausage Digit)

- Dactylitis is very characteristic of SpA although not entirely specific. It is not as common in AS but more typical of reactive arthritis (ReA), psoriatic arthritis (PsA) or undifferentiated SpA (DD).
- Unlike synovitis, the swelling is not confined to a joint but involves the whole digit.
- Combination of synovitis, enthesitis, tenosynovitis, soft tissue swelling.

6) Enthesitis

- Painful inflammation of entheses (the sites of attachment of tendons, ligaments, fascia or joint capsules to bone) is the distinguishing pathological feature of SpA.
- The most typical enthesitis is heel pain (posterior or inferior) related to inflammation of the Achilles tendon or the plantar fascia insertion.
- Pain appears in the morning, as soon as the patient sets foot on the floor and then disappears after a few hours.
- Calcific tendinitis.

5. ASAS Criteria of SPA.

6. Spa features

(entry متتساش نكتب ال)

✓ To be applied in patients with chronic back pain and age at onset of back pain <45 years.

[≥ 1 Spa feature* + Sacroiliitis] OR [HLA-B27 + ≥ 2 other Spa features]

Spa features:

1. Inflammatory spinal pain.
2. Arthritis.
3. Enthesitis.
4. Uveitis.
5. Dactylitis.
6. Psoriasis.
7. Crohn's/colitis
8. Good response to NSAIDs
9. Family history of SpA
10. HLA-B27
11. Elevated CRP

7. Compare between RA & SPA.

غالبا ده السؤال اللي ب 6 درجات ، الدكتور في المراجعة اكدت عليه و قالته نركز هي طالبه ايه كنقط مقارنة ، بعدها قالت 6 نقط مقارنة وسبحان الله السؤال ب 6 درجات (٨-٨) ، النقط هي :-

	Feature	RA	SpA
1	age	child bearing period between the ages of 20-40	young adult often < 45 years for Axial SpA at onset of manifestation
	sex	more in Female	more in male
	genetic	HLA DR4 DR1	HLA B27
2	serology	Rheumatoid factor + anti CCP +	Rheumatoid factor - anti CCP -
3	distribution Synovitis	prominent	can occur but enthesitis + axial involvement are major
4	Distribution	symmetrical peripheral small joint poly arthritis	symmetrical Axial + Asymmetrical oligo articular peripheral arthritis especially LL
	hand	spare DIP	may involve DIP
	enthesitis / Dactylitis	no	occur
5	major non articular manifestation	Rheumatoid nodule	unilateral, symptomatic uveitis
6	treatment	Conventional DMARDs or Biological DMARDs	Biological TNF inhibitor or Biological therapies

3. SLE

8. Laboratory criteria.

Domain	Criteria	Points
anti phospholipid Ab	anti-cardiolipin Ab or anti $\beta 2$ GP1 Ab or lupus anticoagulant	2
Complement proteins	Low C3 or Low C4 Low C3 and low C4	3 4
SLE-specific Ab	anti ds DNA antibody or anti Smith Ab	6

9. Laboratory investigations. (انت عامله ليه او هتلاقي فيه ايه investigation توضح في كل)

Management. (هندي كذا ، وهكذا زي الكتاب يعني lupus nephritis توضح برضو كل حالة هتاخذ كذا ، لو لقينا).

- Complete blood count (CBC) with differential
- Serum creatinine
- Erythrocyte sedimentation rate (ESR) ↑
- C-reactive protein (CRP) ↑ or normal
- Complement levels (C3, C4)
- Liver function tests
- Urinalysis with microscopy
- Protein/creatinine ratio
- 24-hour urine protein

Key Findings in SLE

- Leukopenia, lymphopenia, anemia, thrombocytopenia.
- Elevated ESR with normal or slightly elevated CRP.
- Low complement levels (C3, C4) during disease activity.

Serology tests are crucial for diagnosing SLE and monitoring disease activity:

ANA (Antinuclear Antibody)	Anti-dsDNA	Anti-Sm (Smith)	Other important antibodies in SLE
Screening test for SLE.	More specific for SLE	Most specific antibody for SLE.	• Anti-cardiolipin antibodies. • Anti-$\beta 2$GP1 antibodies. • Lupus anticoagulant.
• High sensitivity (95%). • Not diagnostic without clinical features. • Positive in many other autoimmune conditions.	• High specificity for SLE. • Moderate sensitivity (70%). • Level varies with disease activity. • Associated with lupus nephritis.	• Highest specificity for SLE. • Low sensitivity (30-40%). • Rarely found in other conditions.	

10. Management of SLE ?

pharmacological

- 1) **Antimalarials:** as hydroxychloroquine
 - First-line therapy for most SLE patients.
- 2) **Corticosteroids:** (mild Flare) as methylprednisolone, prednisone
 - For acute flares and maintenance.
 - Short-term use recommended.
- 3) **Nonbiologic DMARDs:**
 - as → cyclophosphamide, methotrexate, azathioprine, mycophenolate, cyclosporine.
 - For moderate to severe disease.
 - immunosuppression :- MAC (mycophenolate, azathioprine, cyclophosphamide)
- 4) **NSAIDs:** For pain and inflammation e.g., ibuprofen, naproxen, diclofenac
- 5) **Biologic DMARDs:**
 - For refractory disease
 - as → belimumab, rituximab, IV immune globulin.

Treatment for Mild Disease

Mild manifestations of SLE often wax and wane with disease activity & can be controlled with:

Cutaneous Manifestations

- Manifestations as → Malar rash, discoid lesions, photosensitivity
- First-line → Hydroxychloroquine, sun protection, topical corticosteroids.
- Second-line → Low-dose oral corticosteroids, methotrexate

Musculoskeletal & Serositis

- Manifestations as → Arthralgia, arthritis, pleuritis, pericarditis.
- First-line → NSAIDs, hydroxychloroquine.
- Second-line → Short course of corticosteroids.

NB: Key Principles → Hydroxychloroquine is the cornerstone of SLE treatment.

Treatment for Severe Disease

- 1) **Lupus Nephritis**
 - Induction therapy → Oral mycophenolate mofetil or cyclophosphamide.
 - Maintenance → Mycophenolate or azathioprine.
 - Biologic therapy → Belimumab or Rituximab. (ملاحظة بخط اليد: B-inhib)
- 2) **Severe Neurologic Involvement**
 - High-dose intravenous steroids.
 - Cyclophosphamide.
 - Rituximab for refractory cases.
- 3) **Systemic Vasculitis:**

High-dose corticosteroids, Cyclophosphamide.



4) Profound Thrombocytopenia

- High-dose corticosteroids.
- Intravenous immunoglobulin.
- Rituximab.

5) Diffuse Alveolar Hemorrhage

- High-dose intravenous steroids.
- Cyclophosphamide.
- Plasmapheresis.
- + rapidly progress glomerulonephritis acute emergency in patient with SLE

4. Osteoarthritis

11. pathology of osteoarthritis & Corresponding radiological findings.

Pathology

- Degeneration changes begin in articular cartilage → Loss of articular cartilage → joint-space narrowing.
- Focal lesion of articular cartilage → uneven loss → Asymmetrical joint space narrowing
- hypertrophic reaction in subchondral bone → ↑ Bone formation → subchondral sclerosis
- marginal cartilage stress → new Bone formation at the joint margin → osteophytes

Imaging in Osteoarthritis

1) Weight-bearing X-rays are mandatory for knee and hip OA. (ملاحظة بخط اليد: mcq)

2) Radiographic hallmarks of "hypertrophic OA" include:

- Localized joint space narrowing >> Indicates cartilage loss.
- Subchondral bone sclerosis >> increased bone density beneath the joint surface.
- Osteophyte formation >> Bony projections at joint margins.
- Bone cysts >> Fluid-filled cavities within the bone.

3) Additional Imaging: MRI and ultrasound can be used for certain occasions to provide more detailed assessment of soft tissues and early changes not visible on X-ray.

Final Page

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